2.6 Feasibility Study

The system is technically feasible because its processing rate and limits are explicitly bounded. The browser requests aircraft data every 1 second and status information every 4 seconds. Live records are removed from the active snapshot after 45 seconds without data; simulation records use a 12-second stale limit. A trajectory sample is stored after 5 seconds or when movement exceeds configured distance, altitude, heading or speed thresholds. Prediction uses 4 to 20 recent samples and is limited to a maximum outage age of 180 seconds. These values are implemented configuration limits rather than claimed aviation-performance guarantees.

The selected software stack is suitable for a single-host academic deployment. Python and Flask provide the application and API layer [3], [4]; SQLite provides embedded local storage [5] and the project enables WAL mode for efficient local read/write activity [6]. Leaflet and Chart.js provide map and chart rendering [7], [8]. The principal constraints are receiver-data quality, internet access for map tiles and optional enrichment, and reduced prediction accuracy during sharp turns or prolonged outages.

Implemented Technical Thresholds

Parameter	Implemented value	Purpose
Aircraft API refresh	1 s	Updates markers and selected-aircraft data.
Status refresh	4 s	Updates live/simulation and receiver-health badges.
Live stale limit	45 s	Removes aircraft no longer present in the live snapshot.
Replay stale limit	12 s	Keeps deterministic replay responsive.
Storage thresholds	5 s; 30 m; 20 m; 5 deg; 2 m/s	Stores time or meaningful movement changes.
Regression window	4-20 samples	Provides enough points while emphasizing recent motion.
Prediction age limit	180 s	Prevents long, increasingly uncertain extrapolation.
Trajectory retention	30 days	Bounds local historical storage.

2.7 Product Functions

- Select live or replay input, decode and normalize aircraft records.
- Display live and predicted aircraft with clear visual distinction.
- Store, retrieve and export trajectory history and logging status.